

What Does an OCR Model Look At? Probing Implicit Component Discrimination in Deepseek-OCR

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Abstract

Document layout components form a fine-grained visual taxonomy: a single page may contain titles, text blocks, figures, captions, tables, equations, and footnotes. End-to-end vision-language OCR models receive no component-level supervision yet must implicitly discriminate among these categories to transcribe a page. We extract spatial attention from Deepseek-OCR’s vision encoder across OmniDocBench. We define attention efficiency (a component’s attention share divided by its area share) and find a consistent directional split: text-bearing components are over-attended (text blocks $1.10\times$ and table captions $1.17\times$), non-text regions under-attended (figures $0.94\times$ and masks below $0.95\times$). 9 of 19 testable categories deviate significantly from proportional allocation after Bonferroni correction. Per-document correlations between attention density and OCR accuracy are significant for several components. After partial-correlation control, three components retain significant effects: figure density ($r = -0.13$, $p < 0.01$), header density ($r = +0.09$, $p = 0.01$), and title density ($r = -0.06$, $p = 0.04$).

1. Introduction

Fine-grained visual categorization (FGVC) asks how well a system distinguishes visually similar categories that differ in subtle ways. Document pages fit this description: a single page can contain a dozen component types whose visual differences are small.

End-to-end vision-language OCR models like DeepSeek-OCR [12] transcribe pages without receiving explicit labels for these component types. Yet correct transcription requires distinguishing them: a table caption contains prose while table cells contain numbers; a figure

region contains no extractable text while its caption directly below does. The OCR training objective forces the model to solve fine-grained categorization as a latent sub-task. Two questions follow: (1) does the model actually learn an implicit component taxonomy, and (2) does its attention allocation predict recognition quality?

Using DeepSeek-OCR’s multi-resolution mode (a global view at 1024×1024 plus up to 6 local crops at 640×640), we extract attention from all 24 layers of the CLIP ViT-L encoder across 1,355 pages from OmniDocBench [9]. Our contributions:

- (i) Per-component attention efficiency on OmniDocBench for 22 component types, showing a statistically consistent bias toward text-bearing components (9 of 19 categories significant after Bonferroni correction, with directional consistency, meaning that every significant text-bearing (apart from page numbers) category has *efficiency* > 1.00 , while every significant non-text category has *efficiency* < 1.00).
- (ii) A complexity-controlled correlation analysis showing that most raw attention-accuracy correlations are confounded by document complexity, but three components (figure, header, title) retain significant partial correlations after controlling for character count and number of component types.

2. Related Work

Donut [5], Nougat [1], Pix2Struct [7], GOT-OCR [11], and DeepSeek-OCR [12] encode document pages as visual tokens and decode structured text. A parallel line of work trains *supervised* layout detectors on component-labeled data such as PubLayNet [15] and DocLayNet [10]. We instead ask whether end-to-end OCR models acquire an analogous component taxonomy *without* component-level su-

pervision, and to our knowledge none of the above models have been probed for how their encoders distribute spatial attention across component types.

OmniDocBench [9] provides 1,355 evaluation pages (v1.5 subset) with 22 component types and per-page OCR ground truth. Prior work uses it only for accuracy benchmarking; we repurpose its spatial annotations as ground truth for attention analysis.

3. Method

Whether attention faithfully explains model behavior is debated [3, 4, 13]. We treat attention as a probe of capacity allocation, not a causal explanation. Self-supervised ViTs have been shown to develop attention patterns that recover semantic structure without explicit supervision [2], motivating an analogous probe of DeepSeek-OCR’s encoder over document components.

Our method quantifies the spatial attention allocation of DeepSeek-OCR. The Gundam preprocessing produces two views. **Global View:** The full document is processed at a base resolution (1024×1024), passes through a SAM-base encoder [6] and a 16× compressor producing a coarse (16×16 grid, 256 tokens). **Local View:** The full image is divided into N local crops (up to $N = 6$) at 640×640 pixels, each producing a 10×10 grid (100 tokens). The number of crops N is determined by the document’s aspect ratio, with a maximum of $N = 6$ for typical 3:4 documents.

3.1. Attention Capture

We extract self-attention weights from the 24-layer vision encoder using forward hooks. For each layer L and attention head H , we compute the attention weight matrix.

$$\mathbf{A}_{l,h} = \text{Softmax}\left(\frac{\mathbf{Q}_{l,h}\mathbf{K}_{l,h}^\top}{\sqrt{d_k}}\right) \in \mathbb{R}^{S \times S} \quad (1)$$

where S is the sequence length (256 patches + 1 CLS token for global view; 100 patches + 1 CLS for local crops).

We measure “attention received” by taking the column-wise mean of the attention matrix, then averaging across all heads H , then computing the mean of 24 layers:

$$M_i = \frac{1}{L \cdot H} \sum_{l=1}^L \sum_{h=1}^H \left(\frac{1}{S} \sum_{j=1}^S A_{l,h}[j, i] \right) \quad (2)$$

$$\bar{\mathbf{A}} = \frac{1}{L} \sum_{l=1}^L \mathbf{A}_l \quad (3)$$

We then extract the CLS token row and reshape to a 2D grid. We zero the first two patch positions per view to suppress attention-sink artifacts [8, 14] (Appendix B).

We quantify efficiency by intersecting $\mathbf{M}_{\text{fine}}$ with ground-truth boxes. We apply a majority-rule threshold,

where a sub-cell (i, j) is labeled if at least 50% of its area is intersected by a bounding box. For each semantic category c , we define attention fraction A_c as the fraction of total attention mass falling on subcells assigned to c , and area fraction F_c as the fraction of total subcells assigned to c . Efficiency is then $D_c = A_c/F_c$. A value $D_c < 1.0$ indicates under-attention, while $D_c > 1.0$ indicates over-attention.

3.2. Significance Testing

We evaluate whether observed attention density deviates from uniform allocation ($D_c = 1.0$) across $N = 1355$ documents. Given that attention distributions in complex documents exhibit high skews, we apply:

- **One-sample t -test** to evaluate the mean density against the null hypothesis $H_0 : D = 1.0$.
- **Wilcoxon signed-rank test** to evaluate whether the median density is significantly shifted from 1.0: $H_0 : \text{median}(D - 1) = 0$, $H_1 : \text{median}(D - 1) \neq 0$.

Bonferroni Correction. To control the family-wise error rate across 22 semantic categories, we set $\alpha_{\text{corrected}} = 0.05/22$. Only results with $p < \alpha_{\text{corrected}}$ are statistically significant. While p -values determine whether an effect is detectable, the effect is not quantifiable. To assess significance, we report effect size measures for parametric (Cohen’s d) and non-parametric (Rank-Biserial) tests. Per-category density-vs-accuracy correlations are reported as Pearson r and Spearman ρ (Appendix C).

4. Experiments

4.1. Dataset

We evaluate on OmniDocBench, a document parsing benchmark containing 1,355 page-level images from diverse sources. Ground-truth annotations provide 8-point polygon segmentations for semantic regions, grouped into 22 categories (see Figure 2 for the complete list).

4.2. Model

We analyze attention in DeepSeek-OCR Gundam, which combines:

- **Vision encoder:** SAM-base (80M) + CLIP ViT-Large (300M) with 24 layers
- **Decoder:** DeepSeek-3B-MoE (30 layers)

5. Results

5.1. Component-level attention efficiency

Figure 2 shows our overall results; full numerical values appear in Appendix D. The largest positive deviations are on dense text regions: table captions ($1.17\times$, $d=+0.77$), text blocks ($1.10\times$, $d=+0.76$), figure captions ($1.06\times$, $d=+0.29$). The largest negative deviations are on regions

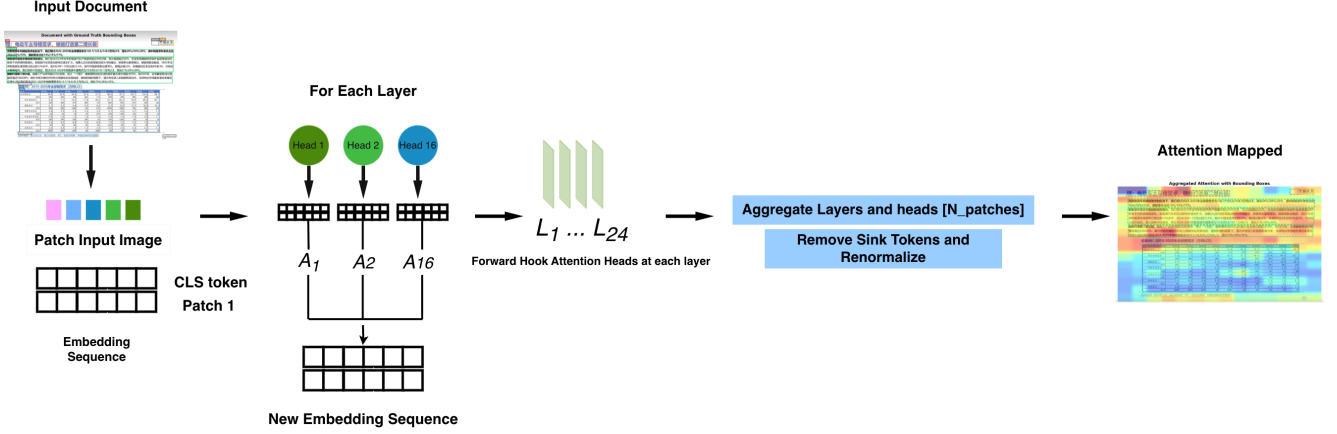


Figure 1. Attention extraction pipeline for the global view (1024×1024 , 256 tokens). An input document is divided into patches and embedded into a token sequence containing a CLS token and 256 patch tokens. For each of the 24 ViT layers, self-attention matrices are intercepted via forward hooks across all heads. Prior to aggregation, the first $n_{\text{sink}}=2$ patch positions are zeroed and renormalized. Attention weights are then averaged across all heads and layers to produce a scalar attention value per patch, reshaped into a 2D spatial grid. Warmer colors in the final heatmap indicate higher received attention; cooler colors indicate lower attention.

with little readable text: figures ($0.94 \times$, $d=-0.60$), table masks ($0.92 \times$, $d=-0.86$), abandoned regions ($0.93 \times$, $d=-0.26$). Tables sit at exactly $1.00 \times$ (not significant), which is expected: they contain both structure and readable text, pulling in opposite directions.

After three categories were excluded due to having less than 10 documents, 9 of the remaining 19 testable categories are significant after Bonferroni correction. Under the null hypothesis of no component discrimination, we would expect roughly one false positive at this threshold; the observed nine, with directional consistency, is well beyond chance. The ten categories that do not reach significance are either genuinely mixed (tables contain both structure and text, averaging out at $\eta=1.00$), or near the noise floor (titles at $1.02 \times$ with $d=0.09$; headers at $0.98 \times$ with $d=-0.08$).

5.2. Attention density vs. OCR accuracy

The efficiency split in §5.1 shows the encoder discriminates component types, raising the question: does the attention density a document allocates to a given component predict how accurately it gets transcribed?

Raw correlations suggest yes for several components (Figure 3). Per-document text_block density correlates positively with accuracy ($r = +0.12$, $n = 1,239$, $p < 0.001$); figure density correlates negatively ($r = -0.11$, $n = 543$, $p < 0.01$). Aggregate content efficiency (attention on all content types divided by area share) also correlates with accuracy ($r = +0.08$, $p = 0.003$). The sign structure matches the efficiency analysis: components that are over-attended are the ones whose density appears to help, and vice versa.

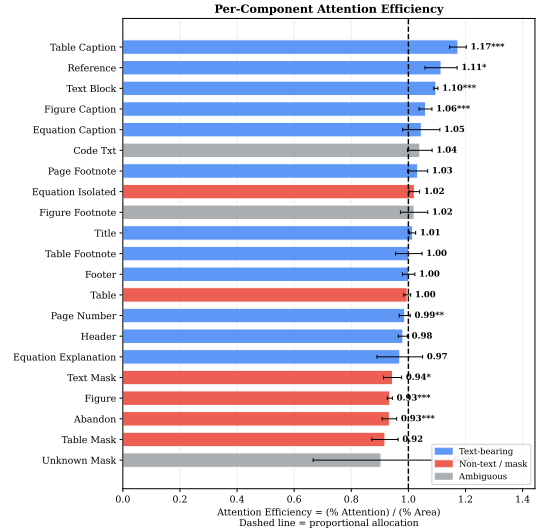


Figure 2. Per-component attention efficiency. Blue bars are text-bearing, red are non-text/mask, grey are ambiguous. Dashed line marks proportional allocation. Stars are Bonferroni-corrected. 21 bars from three categories excluded for having less than 10 samples but includes code.txt and unknown.mask.

5.3. Controlling for document complexity

The raw correlations above have an obvious confound. Complex documents are harder for OCR and spread attention more thinly. Simpler documents may therefore have both higher per-component density and higher accuracy without any direct link between the two. We test this by

Per-Component Correlations Between Attention Density and OCR Accuracy

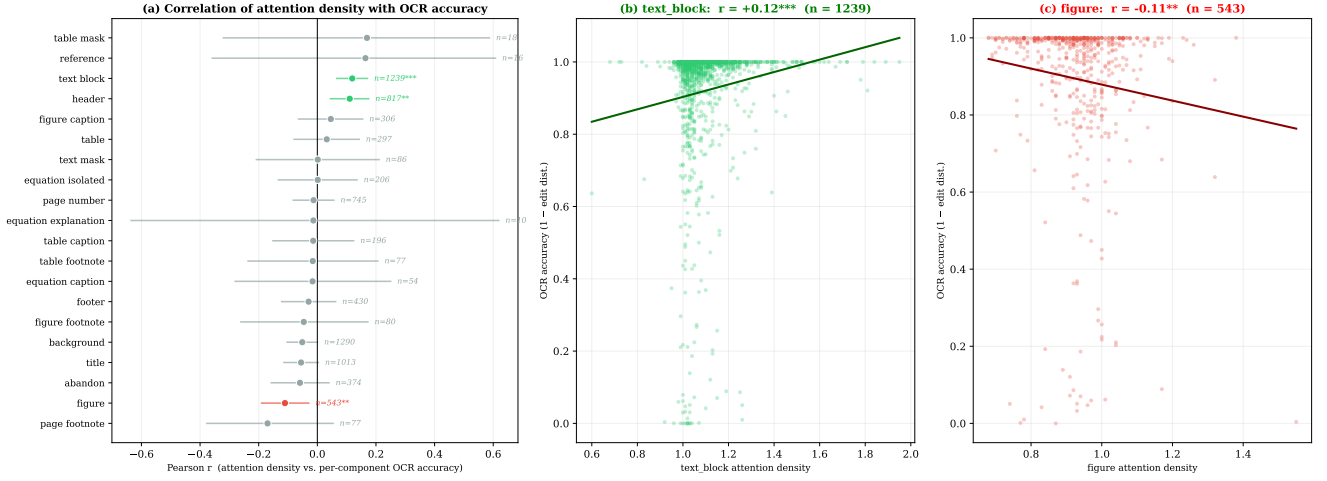


Figure 3. (a) Forest plot of raw Pearson r between per-component attention density and OCR accuracy, with 95% Fisher- z confidence intervals. Green = significant positive ($p < 0.05$), red = significant negative, grey = n.s. (b–c) Per-document scatter plots for text_block ($r = +0.12$) and figure ($r = -0.11$).

Table 1. Attention density vs. OCR accuracy before and after controlling for character count and number of component types.

Component	n	r_{raw}	r_{partial}	p_{partial}
header	817	+0.11**	+0.09**	0.01
text_block	1,239	+0.12***	+0.02	0.41
table	297	+0.03	-0.01	0.86
table_caption	196	-0.01	-0.06	0.42
title	1,013	-0.06	-0.06*	0.04
figure	543	-0.11**	-0.13**	<0.01

computing partial correlations, controlling for ground-truth character count and number of component types present per page. These two variables were chosen as the most direct proxies: character count captures information density and component type count captures layout complexity, without introducing collinearity from redundant spatial measures.

Table 1 shows most raw correlations are attenuated or eliminated. The text_block signal drops from $r = +0.12$ to $r = +0.02$ ($p = 0.41$), and the aggregate content-efficiency correlation drops from $r = +0.08$ to $r = -0.01$ ($p = 0.77$).

Three components show residual effects that resist the confound. Figure density retains a negative partial correlation ($r = -0.13$, $p < 0.01$, $n = 543$). Header density retains a positive partial correlation ($r = +0.09$, $p = 0.01$, $n = 817$). Title density shows a small negative partial effect ($r = -0.06$, $p = 0.04$, $n = 1,013$). All three effects are small ($|r| \leq 0.13$). Header density shows a mean efficiency near 1.0, but across-document variance in header density carries predictive signal after complexity control.

6. Discussion

Implicit taxonomy. The encoder has learned a fine-grained component taxonomy without component-level supervision. Attention probing recovers this taxonomy cleanly, making document components an attractive FGVC testbed: a stable taxonomy of 20+ visually similar categories, ground-truth annotations at scale (OmniDocBench), and an end-to-end task with per-category accuracy.

Complexity as confound. The dominant finding of our correlation analysis is negative: document complexity explains apparent links between attention density and OCR accuracy. We suggest future hypotheses for surviving residuals. For figures, negative partial correlation points to competition between non-text and text recognition. The positive correlation of headers raises the possibility that attention to structural markers proxies for better overall routing. Future work should move from observation to causal intervention, amplifying attention to specific component types.

Limitations. Roughly half the encoder’s attention goes to background regions. Whether reallocating tokens to content would improve accuracy is an open question requiring controlled intervention. We use one model on one benchmark with a uniform mean over all layers and heads. Whether the same bias holds for specific aggregations or on other OCR architectures is left to future work. Our analysis is limited to the encoder; decoder cross-attention may carry other information about component routing. The partial correlations we report are small ($|r| \leq 0.13$). While statistically significant, their effect sizes are modest.

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A. Composite attention construction

For documents requiring local crops, we construct a composite attention map. Each local crop’s 10×10 attention grid is tiled into its corresponding spatial position. The global 16×16 attention grid is bilinearly upsampled to match the composite resolution. Both maps are ℓ_1 -normalized, then blended proportional to token count:

$$\mathbf{M} = \frac{T_g}{T} \hat{\mathbf{G}}' + \frac{T_c}{T} \hat{\mathbf{C}} \quad (4)$$

where $T_g=256$, T_c is the total crop tokens, $T=T_g+T_c$, and hats denote ℓ_1 -normalization. For 6 crops this gives weights ≈ 0.30 global, ≈ 0.70 crop. The composite is renormalized to sum to 1. For a 3:4 document with 6 crops, the composite grid is $30 \times 20 = 600$ cells.

B. Attention sink mitigation

Attention sinks. Xiao *et al.* [14] identify attention sinks—initial tokens that absorb disproportionate attention regardless of content—in autoregressive LMs. Follow-up work extends the phenomenon to ViTs [8]. We zero the first two patch tokens per view and renormalize to prevent sink artifacts from inflating corner regions.

C. Statistical definitions

Cohen’s d .

$$d = \frac{\bar{D} - 1}{s_D} \quad (5)$$

where d is Cohen’s d effect size, $\bar{D}$ is the sample mean of the attention efficiency ratios, 1 is the null-hypothesis value (proportional allocation), and s_D is the sample standard deviation of the attention efficiency ratios.

Rank-biserial correlation.

$$barr = 1 - \frac{2W}{n(n+1)} \quad (6)$$

where $\bar{r}$ is the rank-biserial correlation coefficient, W is the Wilcoxon signed-rank test statistic, and n is the number of observations.

D. Per-component efficiency table

Table 2 reports the full numerical values underlying Figure 2, including sample size, attention mass percentage, efficiency η , Cohen’s d , and rank-biserial $\bar{r}$ for each component.

OCR accuracy. Per-page accuracy is computed as $\text{acc} = 1 - \frac{\text{Levenshtein}(\hat{y}, y)}{\max(|\hat{y}|, |y|)}$ where $\hat{y}$ is predicted text and y is ground truth.

Table 2. Attention efficiency by component type. Bonferroni-corrected significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Component	<i>n</i>	Mass%	$\bar{\eta}$	<i>d</i>	$\bar{r}$
<i>Over-attended</i>					
table_caption	230	0.19	1.17	+0.77***	0.76
text_block	1,239	31.28	1.10	+0.76***	0.92
reference	16	0.35	1.11	+1.08*	0.97
figure_caption	316	0.48	1.06	+0.29***	0.32
<i>Under-attended</i>					
table_mask	19	0.33	0.92	−0.86*	0.80
figure	557	6.61	0.94	−0.60***	0.67
text_mask	88	0.30	0.95	−0.37*	0.43
abandon	399	0.58	0.93	−0.26***	0.36
page_number	788	0.04	0.99	−0.05**	0.16
<i>Not significant (selected)</i>					
table	351	5.81	1.00	−0.04	0.08
title	1,013	2.38	1.02	+0.09	0.04
eq. isolated	207	1.23	1.02	+0.16	0.08

E. Partial correlation scatter plots

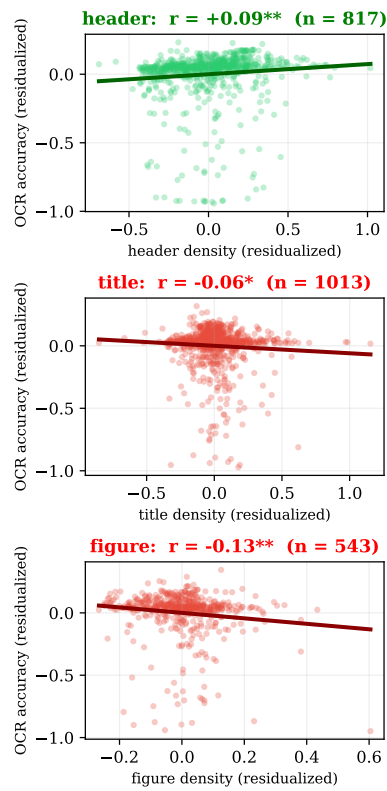


Figure 4. Per-document scatter plots for the three components whose partial correlations survive complexity control ($p < 0.05$). Axes are residualized (character count and number of component types partialled out). Header density predicts better OCR ($r = +0.09$); figure and title density predict worse OCR ($r = -0.13$ and $r = -0.06$ respectively).